SHETH L.U.J. AND SIR M.V. COLLEGE OF ARTS, SCIENCE AND COMMERCE

SUBJECT : ARTIFICIAL INTELLIGECE

PRACTICAL - 3

AIM : Decision Tree Learning

1.Implement the Decision Tree Learning algorithm to build a decision tree for a given dataset.
2.Evaluate the accuracy and effectiveness of the decision tree on test data.
3.Visualize and interpret the generated decision tree.

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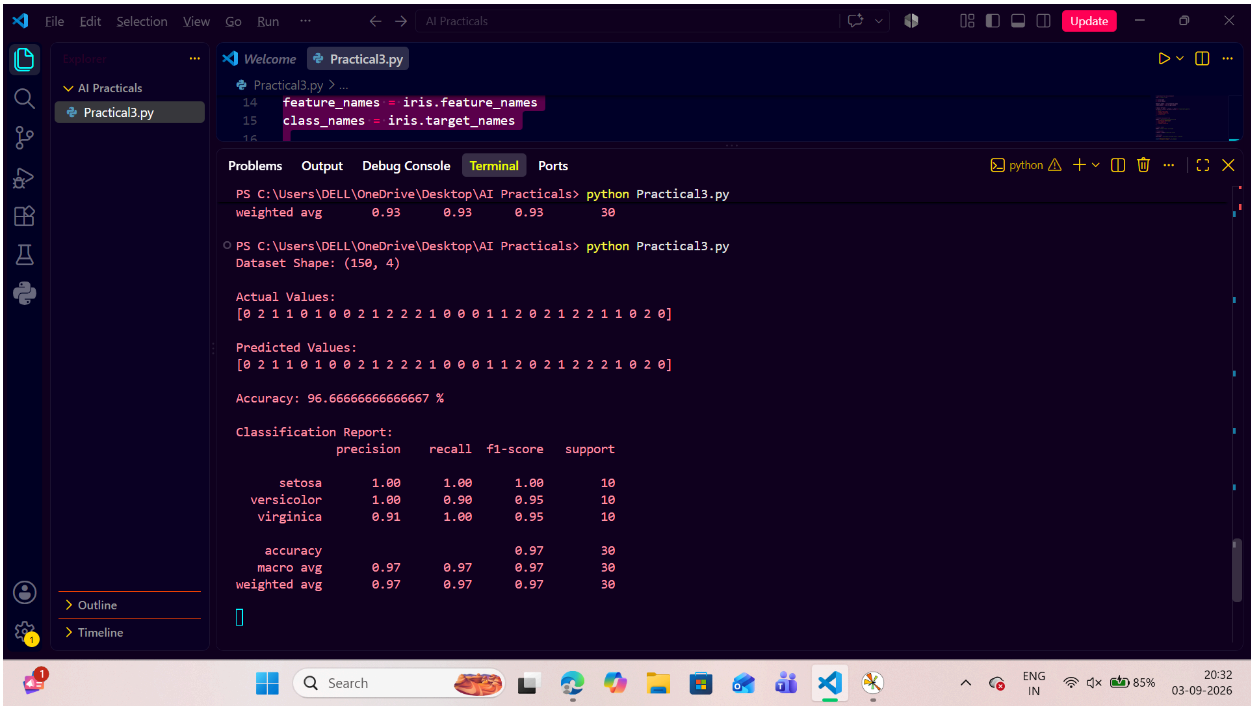
CODE :

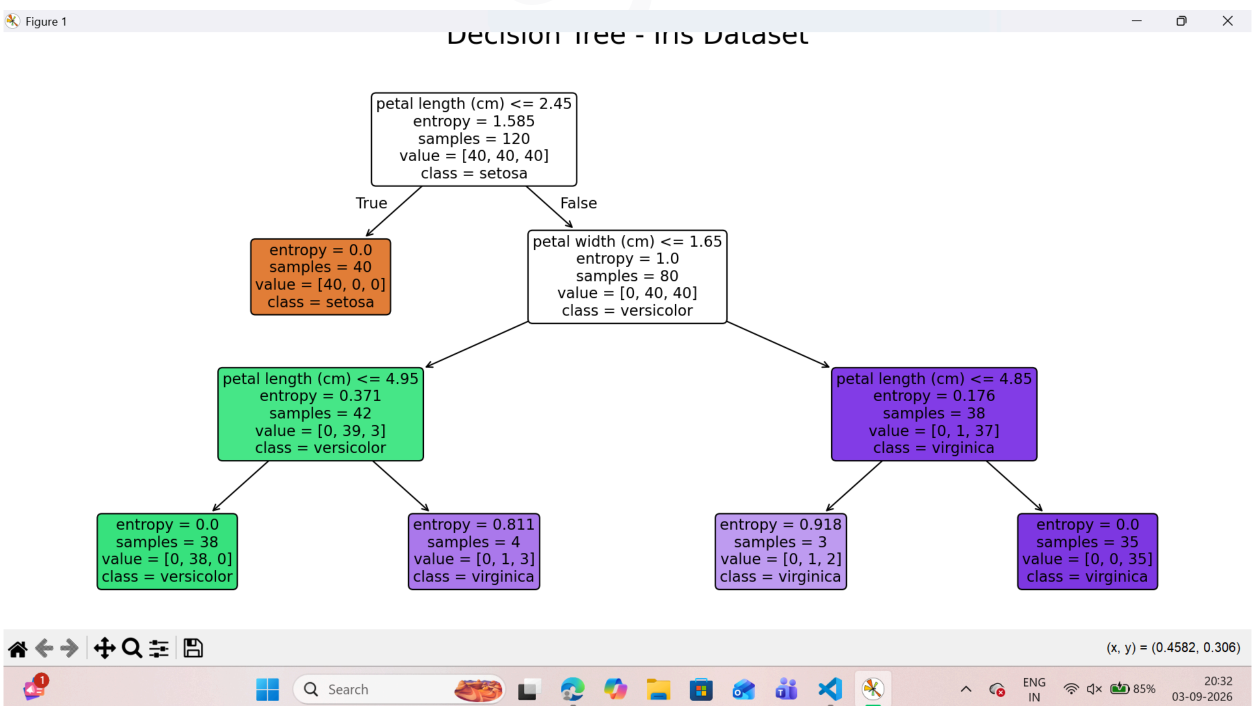
```
import matplotlib.pyplot as plt

from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.metrics import accuracy_score, classification_report

# Load built-in Iris dataset
iris = load_iris()

X = iris.data
y = iris.target

feature_names = iris.feature_names
class_names = iris.target_names

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.20,
    random_state=42,
    stratify=y
)

# Create Decision Tree
model = DecisionTreeClassifier(
    criterion="entropy",
    max_depth=3,
    random_state=42
)

# Train model
model.fit(X_train, y_train)

# Prediction
y_pred = model.predict(X_test)

# Accuracy
```

```
accuracy = accuracy_score(y_test, y_pred)

print("Dataset Shape:", X.shape)

print("\nActual Values:")
print(y_test)

print("\nPredicted Values:")
print(y_pred)

print("\nAccuracy:", accuracy * 100, "%")

# Classification Report
print("\nClassification Report:")
print(
    classification_report(
        y_test,
        y_pred,
        target_names=class_names
    )
)

# -----------------------------
# Decision Tree Visualization
# -----------------------------

plt.figure(figsize=(18, 10))

plot_tree(
    model,
    feature_names=feature_names,
    class_names=class_names,
    filled=True,
    rounded=True,
    fontsize=11
)

plt.title(
    "Decision Tree - Iris Dataset",
    fontsize=20,
    pad=20
)

plt.tight_layout()
plt.show()
```

OUTPUT :



Decision Tree :



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